

# Wicaksono Leksono Muhamad

AI / Machine Learning Engineer & Researcher

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## EDUCATION

### Universitas Gadjah Mada

*Master of Artificial Intelligence*

Yogyakarta, Indonesia

Feb. 2024 – Oct. 2025

### Institut Teknologi Sepuluh Nopember

*B.A.Sc. Computational and Applied Mathematics*

Surabaya, Indonesia

Aug. 2018 – Nov. 2022

## EXPERIENCE

### Research Assistant, Software Engineering Lab

*Universitas Gadjah Mada*

Mar. 2025 – Oct. 2025

Yogyakarta, Indonesia

- Built LLM-based tooling for requirements engineering; added pragmatic and semantic testing on user story quality specifications, extending prior automated QUS analysis.
- Led ML workstream for Laptop Merah Putih grant; built PHQ-aligned LLM screening with personalization and facial affect analysis, evaluated on 302 samples.

### Research Assistant, Machine Learning System

*Universitas Gadjah Mada*

Feb. 2026 – Present

Yogyakarta, Indonesia

- Proposed early-exit training and inference method for energy and latency constrained deployment.

### Researcher, Natural Language Processing

*Mantera Studio Research Lab*

Feb. 2022 – Present

Jakarta, Indonesia

- Published first-author paper at SemEval-2026 Task 11; achieved top-5 rankings across all subtasks on multilingual syllogistic reasoning with 93% accuracy via deterministic parsing.
- Co-authored BEA 2026 Shared Task submission on multilingual lexical complexity prediction; built transformer ensemble with contrastive regression objectives across Spanish, German, and Mandarin Chinese.

### Web Developer

*Mastej Studio*

2019 – 2022

Malang, Indonesia

- Built *What's On Malang* (React/Vite + custom CMS); grew to 400+ daily active users and 100 new leads per day by digitalizing tourism content for the Malang region.
- Delivered Begawan Apartment marketing site with 360-degree virtual tour; generated 10 daily inquiries from outside the Malang region, expanding regional marketing reach.

### Head of Student Resource Development

*Institut Teknologi Sepuluh Nopember*

Feb. 2020 – Nov. 2021

Surabaya, Indonesia

- Led 12 programs across the Student Association; achieved 97% KPI attainment against a 75% target.
- Delivered campus-scale events (MABACUP, UKM EXPO, Managerial School for Event Development) supporting 500+ participants.

## PUBLICATIONS & RESEARCH

### Aligning Implied Statements for Implicit Hate Speech Generalizability

*First Author – Under Peer Review: ACL Rolling Review (OpenReview)*

Jan. 2026

- Trained triplet objective with context-bounded semi-hard negatives to separate near-surface pairs in hate speech embeddings.
- Improved out-of-domain F1-score by 4–8% across 3 hate speech benchmarks over contrastive learning baselines.

### Context-Aligned Contrastive Regression for Multilingual Lexical Complexity Prediction

*First Author – ACL-BEA 2026 Shared Task (arXiv:2605.08950)*

May 2026

- Proposed Context-Aligned Contrastive Regression: Ridge regression ensemble combined with Cross-View Context and Ordinal Soft Contrastive Learning objectives for vocabulary difficulty prediction targeting English learners.
- Achieved top-15 performance across all L1 groups on the BEA 2026 Shared Task benchmark; ranked 14th (Spanish), 11th (German), and 7th (Mandarin Chinese).

## ITLC at SemEval-2026 Task 11: Normalization and Deterministic Parsing for Formal Reasoning in LLMs Mar. 2026

*First Author – ACL-SemEval-2026 (arXiv:2603.02676)*

- Introduced structural abstraction method that transforms syllogisms into canonical logical representations with deterministic parsing to reduce content effects in multilingual reasoning.
- Achieved top-5 rankings of 1200+ submission across all subtasks on the SemEval-2026 Task 11 multilingual benchmark, outperforming complex fine-tuning and activation-level interventions.

## LLM-Based Conversational PHQ-Aligned Screening (CPAS) Mar. 2025 – Dec. 2025

*Person in Charge (ML) – Under Review: IEEE Access; ACM TiiS*

- Designed LLM-driven screening protocol aligned with PHQ-9 combining open-ended dialogue, questionnaire, and facial affect analysis.
- Validated against PHQ reference scores; achieved 0.6–0.8 Spearman’s  $\rho$  correlation and 0.88 agreement on 151 samples.

## Automated Detection of Semantic and Pragmatic Quality Issues in User Stories Mar. 2025 – Jul. 2025

*Research Assistant – Presented: ICODSE 2025*

- Extended AQUASA framework with LLM-assisted checks for user story quality; introduced pragmatic and semantic testing capabilities absent in prior work.

## On-Edge Device Optimization for Audio Classification 2025

*First Author – Published: IEEE Xplore*

- Deployed MFCC-based audio classifier on Arduino Nano 33 BLE; achieved 95.2% accuracy with 3.00 ms latency and 3.80 KB memory footprint.

## Knowledge Distillation for GCN-Based Neural Networks 2022

*Bachelor Thesis – Traffic Volume Prediction*

- Achieved 50% model compression using 3 KD methods (feature-based via adapter, soft-loss, weight transfer) on ASTGCN; improved smaller model accuracy from 30% to 60% against 80% teacher baseline.

## Multimodal Image–Table Pipeline Jun. 2025

*Multimodal Document Pipeline – Paper in Preparation*

- Built document splitting and alignment pipeline for images, tables, and text using graph-based heuristics for structured extraction.

## SOFTWARE PROJECTS

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### Mental Health Screening Platform | *Flask, LangChain, PyTorch, RPC* | mentalhealth.laptopmerahputih.id Nov. 2025

- Shipped PHQ + CPAS screening workflows; served 40+ concurrent user sessions with stable uptime.
- Integrated facial affect capture; reduced p95 latency and cost by 90% through local response batching and request queuing.

### IndoClimate Chatbot | *React, LangChain, RAG* | chat.indoclimate.id Aug. 2025

- Developed WhatsApp webhook + API with RAG over Indonesian climate change legislation for UGM x Deakin University; built LLM summarization pipeline for long legal documents to improve similarity matching.

### TASYA Academic Chatbot | *Flask, LangChain* | tasya.dtmi.ugm.ac.id Feb. 2025

- Built WhatsApp webhook + API for academic support RAG; developed LLM + heuristic document parsing pipeline for multimodal university documents.

### Security Operations Centre Dashboard | *Vite.js, Redux Toolkit, HeadlessUI, Go* Aug. 2024

- Refactored frontend into domain-driven design; reduced hardcoded and redundant code by 70%. Integrated server telemetry ingestion and LLM-based summarization.

## TECHNICAL SKILLS

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**Programming Languages:** Python, TypeScript/JavaScript, Go, Java, C/C++, Rust, SQL (PostgreSQL)

**ML / AI Frameworks:** PyTorch, TensorFlow, HuggingFace Transformers/Trainer, scikit-learn, LangChain, data visualization, data analysis

**Web Frameworks:** React, Next.js, Flask, Node.js, Vite.js

**Data & Tools:** Pandas, NumPy, Matplotlib, Docker, Git, Google Cloud Platform, Linux, L<sup>A</sup>T<sub>E</sub>X

**Languages:** Bahasa Indonesia (Native), English (Professional – TOEFL ITP 587), German (B1), Javanese (Conversational)